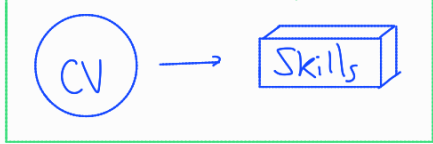
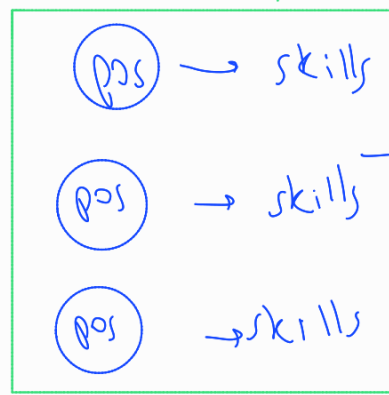


extract struct_data



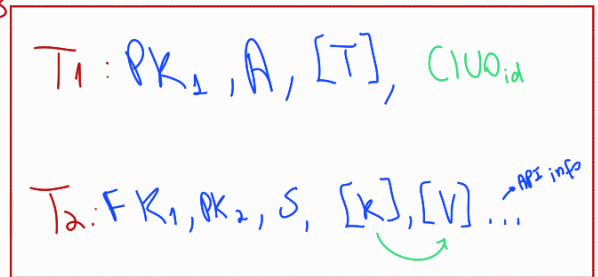
trigger

extract struct_data



CI/O occupations
in the future

VDB



SEARCH LOGIC

10

CV →

A = ENUM (like "TECHNOLOGY")
[T] = main titles (like "product owner", "data analyst", "product manager", "BI analyst")
S = seniority (None)
[K] = hard skills

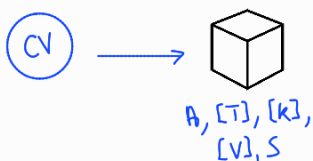
Dict:
K = { name, time of exp }

Kx = missing skills

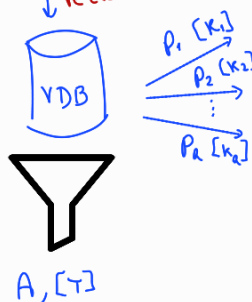
20

Search up VDB for position matches

10: filter only exact A and [T], seniority is optional (may be exact or higher)



API
↓
VECTORIZE



[P_{CV}]

(a) [v₁₁, v₁₂, ..., v_{1m} 0]

(b) [k₁₁, k₁₂, ..., k_{1m} 0]

$$P_0 [K_1, K_2, \dots, K_n]$$

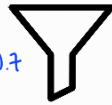
$$P_0 [V_1, V_2, \dots, V_n]$$

(k, v)

cosine similarity

$$\begin{bmatrix} C_{11} & C_{12} & \dots & C_{1(b-1)} & 0 \\ C_{21} & \dots & C_{2p} & \dots & 0 & 0 \\ \vdots & & & & \\ C_{(a-1)1} & C_{(a-1)2} & \dots & C_{(a-1)(b-1)} & C_{(a-1)b} \\ C_{a1} & C_{a2} & C_{a3} & \dots & C_{ab} & 0 \end{bmatrix}$$

$$C_x < 0.7$$



$$C_x \geq 0.7$$



$$\begin{bmatrix} 1 & 0 & \dots & 1 & 1 \\ 0 & \dots & 1 & 1 & 1 \\ \vdots & & & & \\ 0 & 1 & \dots & 1 & 0 \\ 1 & 1 & \dots & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & C_{12} & \dots & C_{1(b-1)} & 0 \\ C_{21} & \dots & 0 & 0 & 0 \\ \vdots & & & & \\ C_{(a-1)1} & 0 & \dots & 0 & C_{(a-1)b} \\ 0 & 0 & \dots & C_{ab} & 0 \end{bmatrix}$$

Use for best potential match positions (not exact)

(x, y)

(x, y)

$$\begin{matrix} (p_1) \\ (p_2) \\ \vdots \\ (p_{n-1}) \\ (p_n) \end{matrix} \begin{bmatrix} V_{11} & 0 & \dots & V_{1(b-1)} & 0 \\ 0 & \dots & V_{2p} & \dots & 0 & 0 \\ \vdots & & & & \\ 0 & V_{(a-1)2} & \dots & V_{a-1} & 0 \\ V_{a1} & V_{a2} & V_{a3} & \dots & 0 & 0 \end{bmatrix}$$

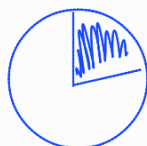
unmatched skills

CLUSTERIZE VECTORS (B clusters)

$$\begin{matrix} V_1 [(i_1, j_1)] \\ V_2 [(i_2, j_2)] \\ \vdots \\ V_b [(i_b, j_b)] \end{matrix} \xrightarrow{(x, y)} \begin{matrix} [K_{x1}, (i_1, j_1)] \\ [K_{x2}, (i_2, j_2)] \\ \vdots \\ [K_{xb}, (i_b, j_b)] \end{matrix} \xrightarrow{\text{identify_skill_and_count}()} \begin{matrix} K_{x1} = t_1 \text{ times } [p_1] \\ K_{x2} = t_2 \text{ times } [p_2] \\ \vdots \\ K_{xb} = t_b \text{ times } [p_b] \end{matrix}$$

$$[K_{xi} = t_i \text{ times } [p_i]]$$

CHARTS



{FEEDBACK}

$$\begin{bmatrix} 0 & K_{12} & \dots & 0 & 0 \\ K_{21} & \dots & 0 & 0 & 0 \\ \vdots & & & & \\ K_{(a-1)1} & 0 & \dots & 0 & K_{(a-1)b} \\ 0 & 0 & \dots & K_{ab} & 0 \end{bmatrix}$$

matched skills

watch out for token consumption and break into steps

$$K + [K_{ij}, i]$$

is_skill_compliant()

$$\begin{bmatrix} 0 & \text{true} & \dots & 0 & 0 \\ \text{true} & \dots & 0 & 0 & 0 \\ \vdots & & & & \\ \text{true} & 0 & \dots & 0 & \text{false} \\ 0 & 0 & \dots & \text{false} & 0 \end{bmatrix}$$

(x, y)

$$K = t \text{ times } [p_i]$$

[Best ^{potential} matching positions]

[Best ^{exact} matching positions]